

Brian Wonjun Lee

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EDUCATION

University of Pennsylvania

BA in Computer Science & Design (Double Major), Benjamin Franklin Scholars

Philadelphia, PA

Expected May 2027

Relevant Coursework: Interactive Computer Graphics, Computer Animation, 3D Computer Modeling, UX & UI Design, Algorithms

EXPERIENCE

Software Engineer Intern

Aleph Lab (Y Combinator)

June 2026 – Present

San Francisco, CA (Remote)

- Shipped a recency-segmented push-notification retention engine (Supabase, Deno, FCM) with an A/B holdout framework to measure return-rate lift
- Built a typed game-AI integration SDK with 4 versioned contract seams (zod, JSON Schema, 128+ tests)
- Resolved ~10 React Native and Next.js defects, including a data-consistency bug traced across mobile, web, and Postgres RPC layers

Software Engineer Intern

BitMango (Puzzle1 Studio)

June 2025 – Aug 2025

Pangyo, South Korea

- Maintained 50+ live Unity mobile game titles, improving UI/UX and core gameplay stability across deployments
- Upgraded SDKs and platform modules to keep titles compliant with evolving app-store requirements
- Resolved 100+ QA-reported bugs by modifying Unity prefabs and C# scripts across live releases

Project Lead & Product Designer

Penn Spark (Red Team)

Jan 2025 – Present

Philadelphia, PA

- Led cross-functional teams of 15+ designers and developers to ship client-facing 0-to-1 products
- Shipped polished, user-centered features across semester-long client project cycles

Military Interpreter

Capital Defense Command (ROK Army)

Dec 2022 – June 2024

Seoul, South Korea

- Provided real-time English-Korean interpretation and translated sensitive briefings in high-security, multi-agency environments

Database Engineer Intern

IT-Farm

June 2022 – Aug 2022

Seongnam, South Korea

- Designed and implemented a SQL database prototype managing 1M+ data entities using Oracle and SQLite
- Processed and organized semiconductor datasets from 20+ client companies, including Samsung, SK, and LG

PROJECTS

[Mini Minecraft](#) | C++, OpenGL, GLSL

Apr 2026

- Generated 7-biome procedural terrain using layered Perlin/FBM/Voronoi noise with smoothstep biome blending
- Implemented 3D Perlin cave systems with trilinear interpolation, fluid physics, and post-process tint overlays
- Built render pipeline with 7x7 PCF shadow mapping, screen-space reflections, vertex AO, and day-night cycle

[Passenger](#) | Unreal Engine 5, HLSL, Maya

Mar 2026 – Apr 2026

- Wrote 2 HLSL post-process shaders (anisotropic Kuwahara filter, cell shader) sampling GBuffer channels for geometry-aware stylization
- Rigged and animated a character in Maya (IK limbs, spline-IK spine) driving an Unreal Animation Blueprint
- Rendered via Movie Render Pipeline with 64 TSR samples per frame at 2560x1080 with hardware ray tracing

[Mini Maya](#) | C++, OpenGL, Qt

Feb 2026 – Mar 2026

- Implemented a half-edge mesh structure with topology-aware editing operations (edge split, extrusion, Catmull-Clark subdivision)

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, C++, C#, SQL, OCaml

Frameworks: React, Next.js, Node.js, React Three Fiber, Tailwind CSS, Astro

Tools: Unreal Engine 5, Unity, OpenGL, GLSL, HLSL, Maya, Blender, Substance Painter, Figma, Git